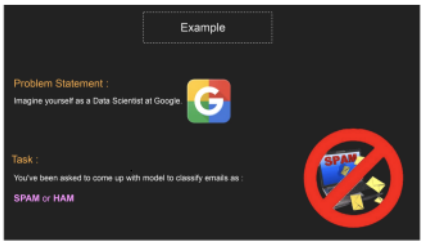
SVM

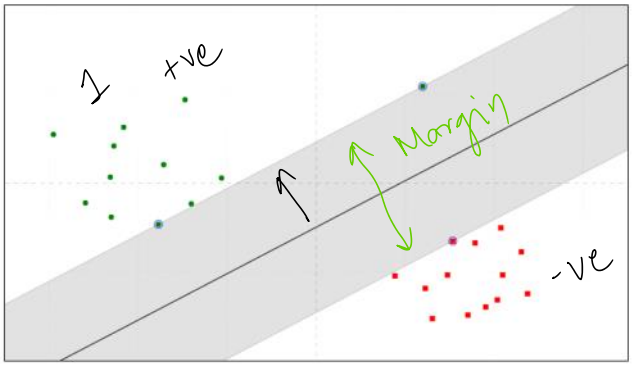
Tuesday, March 12, 2024 9:06 PM

- Linear
- Logistic
- KNN
- DT
- RF
- GBDT
- XGBOOST
- SVM

} Supervised

Classification



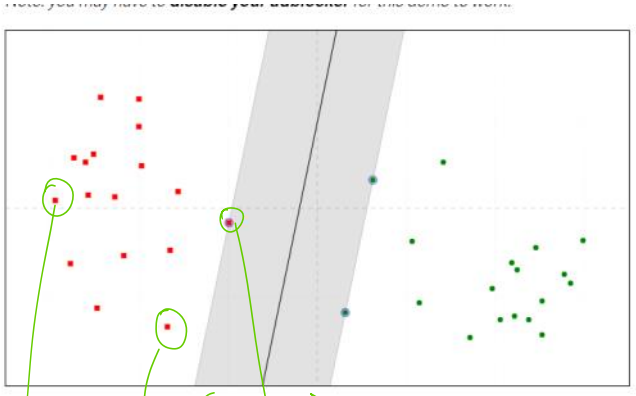


- Find best hyperplane

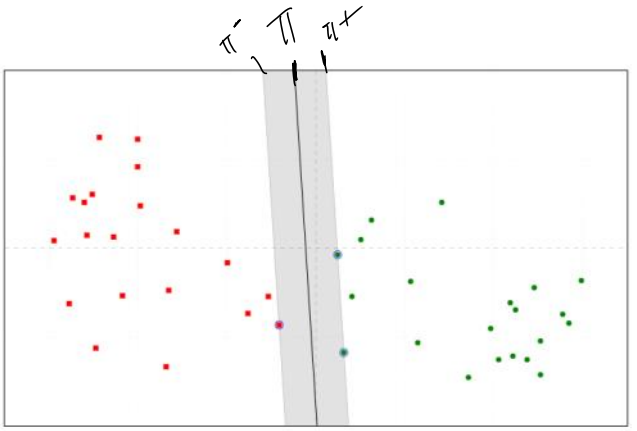
+ve
spam
1

-ve
Non spam
-1

1



2



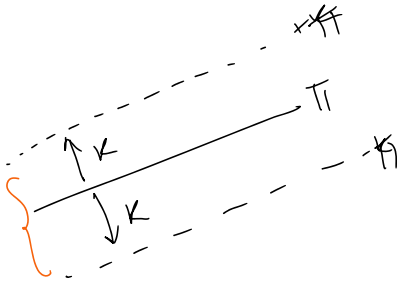
More Margin

Margin Maximing classifier

$w^T x + b = 0 \rightarrow \pi$

$w^T x + b = K$

$w^T x + b = -K$



$\frac{K}{\|w\|}$

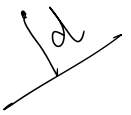
$\frac{K}{\|w\|}$

w_0

+

$w^T x + b - k = 0 \quad \cdot \pi^+$

$w^T x + b + k = 0 \quad \pi^-$



$w^T x + w_0$

$\left\{ \frac{w_0}{\|w\|} \right\}$

$$d(0, \pi^+) = \frac{b-k}{\|w\|}$$

$$d(0, \pi^-) = \frac{b+k}{\|w\|}$$

$$\text{Margin}(\pi^+ \pi^-) = d(0, \pi^-) - d(0, \pi^+)$$

$$= \frac{b+k}{\|w\|} - \frac{b-k}{\|w\|}$$

$$\frac{2k}{\|w\|}$$

—

$$\frac{w_0' - w_0}{\|w\|}$$

distance
||
lines

$$\text{Margin} = \frac{2k}{\|w\|}$$

SVM

find that $\bar{w}$ such that $\frac{2k}{\|w\|}$ is maximum

case ①

$k = 1$

$d = \frac{2 \times 1}{\|w\|} = \frac{2}{\|w\|}$

$\underset{\bar{w}}{\text{argmax}} \left(\frac{2}{\|w\|}\right)$

case ②

$k = 10$

$d = \frac{2 \times 10}{\|w\|} = \frac{20}{\|w\|}$

$\underset{\bar{w}}{\text{argmax}} \left(\frac{20}{\|w\|}\right)$

$\|w\|$	$(2/\|w\|)$	$(20/\|w\|)$
1	2	20
2	1	10
3	0.67	6.8
4	0.5	5
5	0.4	4

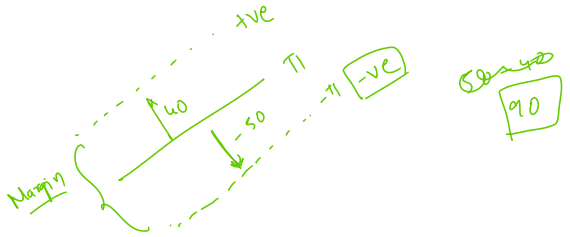
$k = 1$ $\quad [d = \text{margin} = \frac{2}{\|w\|}]$

$\Rightarrow \underset{\bar{w}}{\text{argmax}} \left(\frac{2}{\|w\|}\right)$

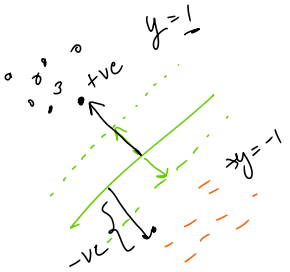
- **π+ : w^T * x + b = 40**
- **π- : w^T * x + b = -50**

then margin will be:

From <https://www.scaler.com/instructor/meetings/i/ml-svm-1-28/>



Hard Margin



$y = 1$ +ve samples

$y = -1$ -ve sample

$y(w^T x + b)$

$y(7)$

1×7

$(7)+$

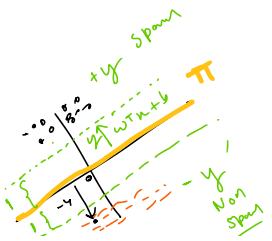
$y(w^T x + b)$

$1 \quad (-7)$

(-7)

Signed distance

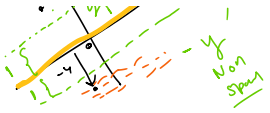
$y(w^T x + b)$



$y = +1 \rightarrow y(w^T x + b) \geq 1$

$1 \quad (-4)$ } ✗ Hard Margin

≥ 1



$1(-4)$ $-4 \geq 1$ } X Hard Margin

$y=-1 \quad y(w^T x+b) \leq -1$

Soft Margin

Margin + error



Max Margin

- Minimize the error

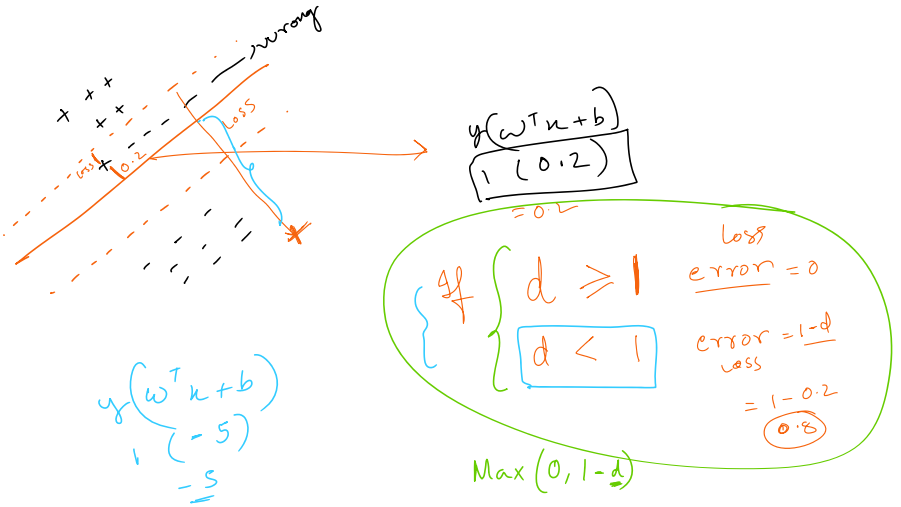
$$\max_{(w,b)} \frac{2}{\|w\|}$$

Reverse

$$\min_{(w,b)} \frac{\|w\|}{2} + C(\text{loss function})$$

Margin / loss } Min

Margin

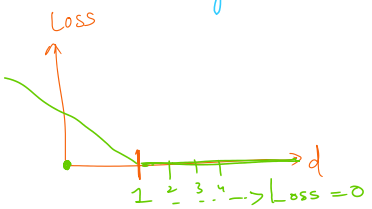


$1-(-5)$

$1+5$

6

Hinge Loss



$$\left\{ \underbrace{\min_{(w,b)} \frac{\|w\|}{2}}_{\text{Margin}} \right\} + C \left(\text{loss function} \right)$$

Loss ≥ 0 } for Misclass

$$\min_{(w,b)} \frac{\|w\|}{2} + \frac{C}{N} \sum_{i=1}^{N} \text{Loss} \quad \text{s.t} \quad (w^T x + b) y \geq 1 - \text{Loss}$$

Loss > 0

$$\text{Max}(0, 1-d)$$

Now, the optimization fn changes to :

$$min_{w,b} \frac{||w||}{2} + \frac{C}{N} \sum_{i=1}^{N} \zeta_i \;\; s.t. \; (w^T x_i + b) y_i \geq 1 - \zeta_i$$

$$\forall\, i : 1 \rightarrow N \; and \; \zeta_i \geq 0$$

This is known as Soft Margin SVM

C → 0 x Misc

C ↓ less → only focus | Margin|

underfit

C → 1000

$$\frac{\|w\|}{2} + C \sum \text{loss}$$

100000 5

100000 ↓↓

2

100000 5

500000

Overfit